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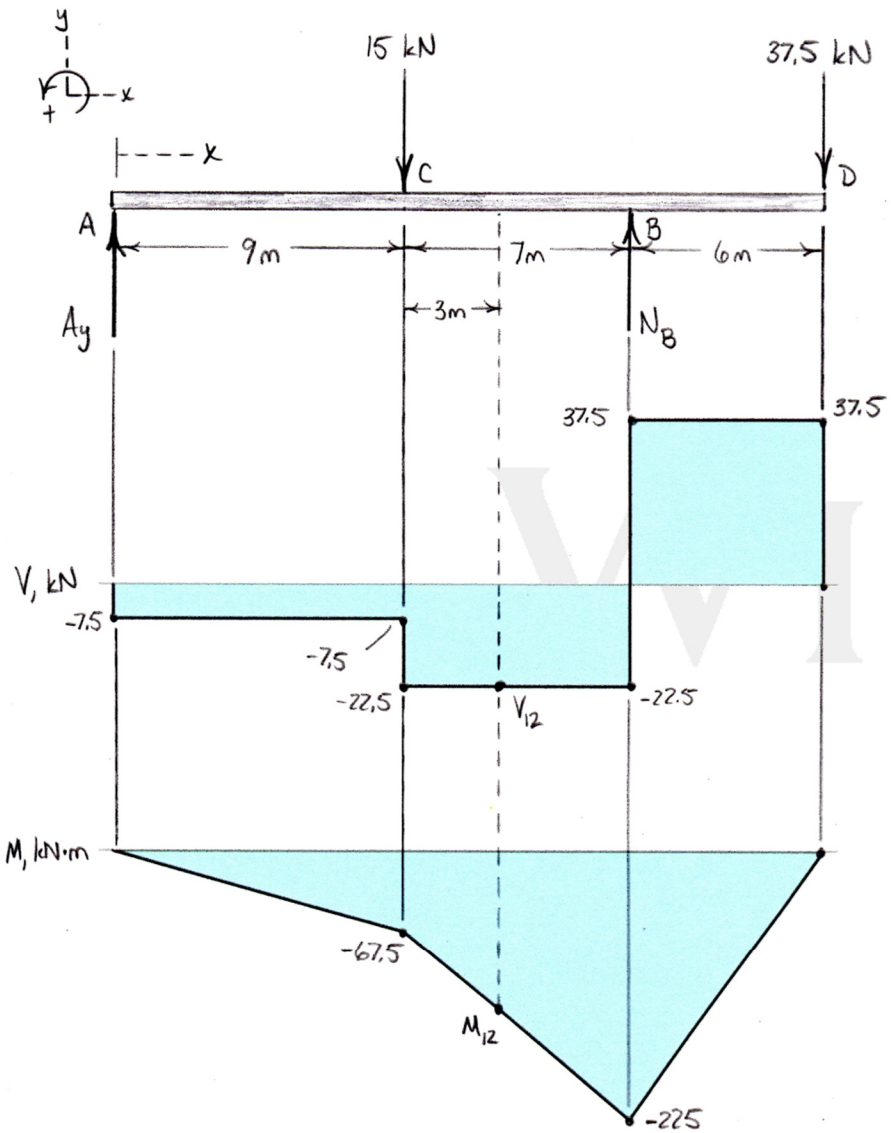

$$\begin{cases} \Sigma F_y = 0: & A_y + N_B - 15 - 37.5 = 0 \\ \Sigma M_A = 0: & -9(15) + 16\,N_B - 22(37.5) = 0 \end{cases} \longrightarrow \begin{cases} A_y = -7.5 \text{ kN } (\downarrow) \\ N_B = 60 \text{ kN} \end{cases}$$

$$\begin{cases} M_C = -7.5(9) = -67.5 \text{ kN·m} \\ M_B = -67.5 - 22.5(7) = -225 \text{ kN·m} \\ M_D = -225 + 37.5(6) = 0 \end{cases}$$

$$\begin{cases} \underline{V_{12} = -22.5 \text{ kN}} \\ M_{12} = -67.5 - 22.5(3) \longrightarrow \underline{M_{12} = -135 \text{ kN·m}} \end{cases}$$